

Master's Comprehensive Examination

Part I

Theory (90 minutes)

March 11, 2000

Instructions: Answer all four questions. Open book and open notes.

1. Let X_1, X_2, X_3 be a random sample from a normal distribution with mean μ and variance 1. Find the MSE for the following two estimators and indicate for what values of μ is one estimator preferred to the other.

$$T_1 = (X_1 + X_2 + X_3)/3 = \bar{X}$$

$$T_2 = (1/5)[(X_1 + X_2 + X_3)/3]$$

2. Let X_1, X_2 be independent Poisson variables with parameter λ .

(a) Find a sufficient statistic for λ .

(b) Which of the following is a sufficient statistic?

(i) $2(X_1 + X_2) + 7$

(ii) $2X_1 + 3X_2 + 1$

3. Let X_1, X_2 be independent with density

$$f_X(x; \beta) = \frac{e^{-x/\beta}}{\beta^2}, \quad x > 0.$$

(a) Find the MLE of β .

(b) Find the MLE of $e^{\beta+1}$.

4. Let X_1, X_2, X_3 be independent normal variables with mean μ and variance σ^2 . Both μ and σ^2 are unknown.

(a) Find an unbiased estimator of $\mu + 2$.

(b) Find an unbiased estimator of $2\mu + 3\sigma^2$.